

EXAMPLE 13**Forward Interest Rate Changes**

At its June 2022 meeting, the US Federal Reserve surprised markets by raising its target interest rate by a greater-than-expected 75 bps and suggested further increases in the future in response to sharply higher inflation. Exhibit 11 shows one-year and two-year US Treasury strip prices per USD100 at the end of May 2022 and after the Fed's decision in June:

Exhibit 11: One- and Two-Year US Treasury Strip Prices

Date	PV(1y)	PV(2y)
31 May 2022	98.028	95.109
15 June 2022	97.402	93.937

- Using the information in Exhibit 11, show the change in the breakeven one-year reinvestment in one year's time ($F_{1,1}$).

Solution:

62 bps or 0.62 percent

First, we use Equation 18 to solve for each market discount rate r . For example, in the case of the two-year discount bond on 15 June:

$$r = 3.177 \text{ percent} = \left(\frac{100}{93.937} \right)^{\frac{1}{2}} - 1.$$

The following table summarizes the respective discount bond rates:

Date	r_1	r_2
31 May 2022	2.012%	2.539%
15 June 2022	2.667%	3.177%

Solve for the respective forward rates ($F_{1,1}$) by rearranging Equation 25:

$$F_{1,1} = (1+r_2)^2/(1+r_1) - 1.$$

$$\underline{F_{1,1}(31 \text{ May})}: F_{1,1} = 3.069\% = (1 + 2.539\%)^2/(1 + 2.012\%) - 1.$$

$$\underline{F_{1,1}(15 \text{ June})}: F_{1,1} = 3.689\% = (1 + 3.177\%)^2/(1 + 2.667\%) - 1.$$

While the higher r_1 and r_2 in mid-June shows that investors have factored the immediate 75 bp increase into their nominal discount rate, the 62 bp (0.62 percent) increase in $F_{1,1}$ is a measure of expected future rate increases given that the one-year nominal rate of return starts in one year's time. Note also that in a rising rate environment, $F_{1,1} > r_2$, as shown in the following diagram comparing r_1 , r_2 and $F_{1,1}$ on 31 May (lower rates) to 15 June (higher rates).